

CS177 Bioinformatics: Software Development and Applications
SIST, ShanghaiTech University

Spring, 2025

Homework 3

Due time: 11:59 pm, May 11, 2025 (Beijing Time).

Submission: In Blackboard, upload your solution in a zip file, including a PDF file with the written solution and Jupyter notebooks with all the outputs, etc.

Note: Please name your zip file as “CS177-HW3-your name-your student ID”, for example, “CS177-HW3-JieZheng-2024233187.zip”.

The questions below are assigned 50 points in total, which will constitute 10% of the final grade.

Q1. Short Answer Questions (10 points)

1. The key steps in AlphaFold2’s protein structure prediction include: an amino acid sequence as input → searching and construction of multiple sequence alignment (MSA) → extraction of co-evolutionary information → prediction of three-dimensional structure through deep learning models → structure optimization. Why is Multiple Sequence Alignment (MSA) crucial for AlphaFold2’s protein structure prediction? (5 points)
2. What are the main pre-training objectives of scGPT? Explain the role of the attention mask in scGPT and how it helps the model perform generative pretraining on non-sequential omics data. (5 points)

Q2. Synthetic Lethality Prediction Using Multi-Modal Embeddings (40 points)

Objective: Build a classifier that predicts **synthetic lethal (SL)** interactions between 2 genes using multiple types of features.

Problem Description: Synthetic lethality (SL) is a type of genetic interaction where the simultaneous inactivation of two genes leads to cell death, while inactivation of either gene alone does not affect cell viability. SL interactions are of great interest in cancer prediction medicine, as they enable the selective targeting of tumor cells based on their unique genetic vulnerabilities.

In this assignment, your task is to predict SL interactions between gene pairs using pretrained embeddings and biological features. You are provided with a training set of gene pairs labeled as SL or non-SL and a separate test set. A mapping file is also provided to convert NCBI Gene IDs to gene symbols. Your model should output a binary label indicating whether each gene pair is synthetic lethal (label = 1) or not (label = 0).

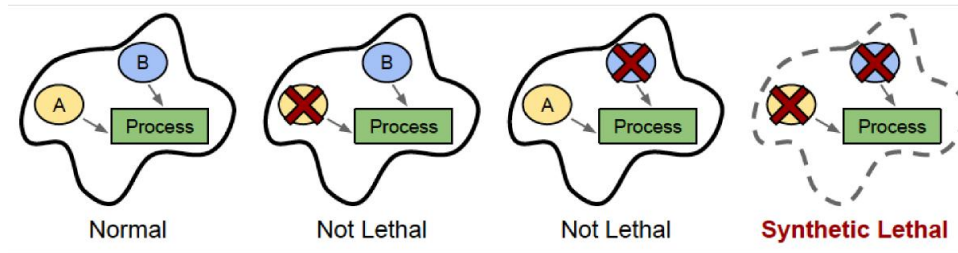


Figure 1: The concept of synthetic lethality.

We provide several types of features for this task:

- **Geneformer Embeddings:** Pretrained embeddings for each gene symbol derived from the Geneformer model. These are provided in `geneformer_gene_embs.pkl`.
- **scGPT Embeddings:** Pretrained embeddings obtained from the scGPT model, representing gene functionality in a single-cell context. These are provided in `scgpt_gene2idx.txt` and `scgpt_emb.pkl`.
- **Protein Features:** Protein sequences and protein-protein interaction (PPI) network data are available for use as additional features. These are provided in `protein_info.csv` and `ppi.csv`.
- **Optional:** You are encouraged to incorporate additional features from external sources if they are expected to improve model performance.

You are expected to experiment with different feature combination strategies and explore various classifiers. If an embedding is missing for a gene in the test set, you should handle it appropriately—for example, by using the average embedding from the training set or applying an imputation method.

Dataset and Evaluation Criteria: The training and test datasets can be downloaded from the Kaggle competition page:

<https://www.kaggle.com/competitions/shanghai-tech-cs-177-hw-3/data>

Model performance will be evaluated using the Area Under the Receiver Operating Characteristic Curve (AUROC). A higher AUROC indicates a more accurate model.

Submission: Submit your predictions for the test set to the Kaggle competition platform:

<https://www.kaggle.com/t/bd94d461bfe14e658ca9d47366264199>

You are allowed up to **five submissions per day**. Select **two final submissions** for evaluation; the one with the higher AUROC will be used for grading. Please upload the final version of your model code and a screenshot of your account on the Kaggle leaderboard (includes your account name) to Blackboard before the deadline.

Scoring: Your total score for this question (out of 40 points) will be distributed as follows:

- **Basic Score (28 points):** Based on the correctness and completeness of your solution, including functional code and valid Kaggle submission.
- **Performance Score (12 points):** If $\text{AUROC} \geq 80\%$, you will get the full points. Computed as $\text{Performance Score} = \min(12, (\text{AUROC} - 0.5) \times 40)$.

END OF CS177 HOMEWORK 3